

IT351 PROJECT - FINAL REQUIRED SECTIONS

Standalone file containing the added final-report requirements

Version History / Commit Log

The following internal record summarizes the main stages represented in the submitted report and Packet Tracer file.

Version	Phase	Work Completed	Status
1.0	Phase 1	Created the initial topology, selected devices, and named routers and switches.	Completed
1.1	Phase 1	Connected LAN devices and prepared the IPv4 addressing plan for all four LANs.	Completed
2.0	Phase 2	Configured LAN interfaces, serial addressing, clocking, and static routes.	Completed
2.1	Phase 2	Verified bidirectional communication with successful ping tests and 0% packet loss.	Completed
2.2	Final	Reviewed commands, added reflection and artifacts, and prepared Word and PDF files.	Completed

Reflection

Completing this project changed the way we approached network troubleshooting because every part of the design had to work as one system. The most significant challenge was establishing reliable communication between the Main Campus and the Branch Campus after the four LANs had been configured. Local communication was straightforward, but the first cross-WAN tests required careful checking of router interfaces, the serial link, default gateways, and routing entries. We reviewed the addressing table, used the router interface status and configuration outputs, and checked the path step by step. The serial connection was enabled on both routers, a clock rate of 64000 was applied on the DCE side, and each required static route was verified. We then repeated the ping tests in both directions. Receiving four replies with 0% packet loss confirmed that the LAN and WAN configurations were working together correctly.

We selected static routing rather than RIP because the topology is small and contains only two routers and four known LANs. Static routes made the forwarding path clear and allowed every destination network and next-hop address to be checked directly. RIP would reduce manual updates if the network grew, but it would introduce unnecessary routing advertisements and additional configuration for this limited design. We also used /24 networks for the LANs because they were easy to organize and distinguish, while the /30 subnet on the serial link provided only the two usable addresses required for the point-to-point connection. The main tradeoff is that static routing is less scalable and would require manual changes if more campuses or networks were added.

AI assistance was used only to proofread the final wording, review document formatting, and identify small presentation issues in the listed commands. It was not used to operate Packet Tracer or replace the group's configuration work. Every technical correction was checked against the actual topology, router outputs, IP settings, and successful ping results before it was included.

If we repeated the project, we would test and document each stage immediately after completing it. We would save clearer screenshots of interface status, routing tables, and ping results, use a structured troubleshooting checklist, and keep more frequent local versions of the report and Packet Tracer file. This would make errors easier to isolate and produce a stronger final submission.

Project Artifacts

Cisco Packet Tracer project file (.pkt): IT351_FinalDesign.pkt (attached separately).

Final written report: IT351_Project-Phase_2_Final.docx.

Final PDF report: IT351_Project-Phase_2_Final.pdf.

Final Submission

Submission Item	Status
Final Word report (.docx)	Prepared
Final PDF report (.pdf)	Prepared
Cisco Packet Tracer project (.pkt)	Attached separately